Question Paper ID: 696162

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Course Code: MA266TEU

USN | 1 | R | V | 2 | 3 | A | I | 0 | 7 | 2 |

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)
R V Vidyanikethan Post
Mysuru Road, Bengaluru - 560059

VI Semester B.E. Regular / Supplementary Examination July / August 2026

(Institutional Elective)

Mathematical Modelling

Time : 3 Hours

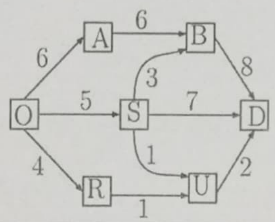
Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Write any two characteristics of mathematical modelling.	02	1	1
1.2	The horizontal distance(range) covered by a soccer ball of mass 2 kg is kicked from the ground with an initial velocity of $30m/s$ at an angle of 30° to the horizontal is __49__. (Take acceleration due to gravity as $g \approx 9.8\,\mathrm{m/s^2}$)	02	2	2
1.3	With respect to the characteristic of brown eyes, suppose the proportions of dominant, hybrid, and recessive individuals in the 9^{th} generation are $p_9 = 0.36, q_9 = 0.48, r_9 = 0.16$. The proportion of recessives in the 10^{th} generation (r_{10}) is ________ and proportion of hybrids in the 11^{th} generation (q_{11}) is________.	02	1	1
1.4	Suppose the population of one-horned rhinoceros in foothills of Himalayas are decreasing at the rate of 2% per year and their population was 400 in 2015, then how many of them will be left approximately in 2024?	02	2	2
1.5	Compute the fixed probability vector of the stochastic matrix $A = \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix}$.	02	1	1
1.6	A travelling salesman **S** sells in 3 cities **A, B, C**. He never sells in the same city on successive days. If he sells in city **A** on a day, then the next day he sells in city **B**. However if he sells in either **B** or **C**, then the next day he is twice as likely to sell in city **A** as in the other city. Find the transition matrix of the chain of selling.	02	2	2
1.7	Draw the bipartite graph $K_{3,5}$ and the complete graph K_4 .	02	1	1

1.8	Express the following transition probability matrix as a weighted digraph. $\begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.25 & 0.35 & 0.4 \end{bmatrix}$	02	2	2
1.9	In the following figure, with respect to dynamic programming, the value of $f_1^*(O)$is __________. 	02	2	2
1.10	The extremal of the functional $\int_0^3 x^3 (y')^2 \, dx$ is __________.	02	1	1

Part B

Question No	Question	M	CO	BT
2a	Model and describe the second-order differential equations representing a mass-spring-dashpot mechanical system. Also highlight the analogies between the elements of the mechanical and electrical systems.	08	3	3
2b	Drug is induced in a patient's bloodstream at regular intervals of duration T each. Simultaneously the drug disappears at a rate proportional to the concentration $c(t)$ of the drug present at any time t. Determine the differential equation governing c(t), solve it and interpret the results.	08	4	4
3a	Discuss the Cobweb Model describing the variation of price and quantity of an agricultural commodity produced by farmers. Using a suitable graph, interpret the behaviour of the model	08	3	3
3b	(i) A machine's value changes every year according to the relation $V_{n+1} = 0.80V_n - 2$, where V_n is the value in lakhs after n years. If the initial value $V_0 = 15$ lakh, find the value of the machine after n years. (ii) Solve the difference equation $2x_{t+2} - 5x_{t+1} + 2x_t = 0$.	08	2	2
	OR			
4a	Model the population dynamics using difference equations and discuss the one period fixed points.	08	3	3
4b	Solve the difference equation $x_{t+2} = 4x_{t+1} - 5x_t$.	08	2	2
5a	An e-commerce website tracks user navigation behaviour between four main sections: The Homepage (H), Product Catalog (C), Shopping Cart (W) and the Payment Gateway (P). From the Homepage (H): Users always navigate directly to the Product Catalog (C). From the Product Catalog (C): There is a 50% chance they stay browsing the Catalog (C) and a 50% chance they add an item to the Shopping Cart (W). From the Shopping Cart (W): There is a 60% chance they return to the Homepage (H), a 20% chance they proceed to the Payment Gateway (P) and a 20% chance they remain in the Cart adjusting	08	4	4

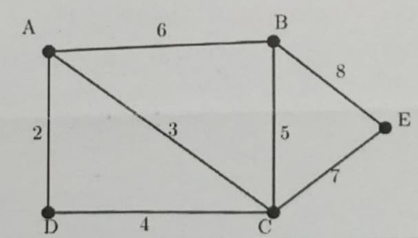
items. From the Payment Gateway (P): After completing or cancelling a transaction, they are always redirected back to the Homepage (H). In the long run, what percentage of a user's active session time is spent on the Payment Gateway (P)?

5b	A company executive changes his bike every year. If he has Pulsar he changes to splendor, if he has Splendor he changes over to Royal Enfield. If he has Royal Enfield, he is just as likely to change over to a bike Royal Enfield, Splendor or Pulsar. If he had Royal Enfield in 2020, find the probability that he will have a bike (i) Pulsar in 2022, (ii) Royal Enfield in 2022.	08	2	2

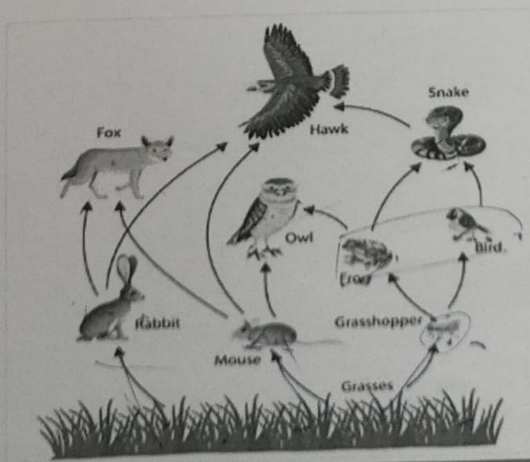
OR

6a	Verify the transition probability matrix $P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ is irreducible or not. Also find its stationary probability vector.	08	2	2
6b	An automated robotic arm on a factory floor moves items between four inspection stations: Receiving (R), Sorting (S), Processing (P) and Re-inspection (M). From Receiving (R): Items always move directly to Sorting (S). From Sorting (S): There is a 40% chance an item stays in Sorting for deeper scanning and a 60% chance it moves to Processing (P). From Processing (P): There is a 70% chance it moves back to Receiving (R), a 10% chance it requires Re-inspection (M) and a 20% chance it stays in Processing (P). From Re-inspection (M): Once checked, it always moves back to Receiving (R). In the long run, what percentage of time does the robotic arm spend in Re-inspection (M)?	08	4	4

7a State the max-flow min-cut theorem. For the network shown below, determine the maximum flow between A and E by enumerating all the A-E cuts and the capacities therein. — 08 4 4

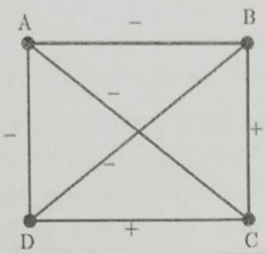


7b Define the Harary measure of status. In the given food web, an edge from X to Y means X is food for Y. Find the status of each species according to the Harary measure. — 08 3 3



OR

8a	Alvin (A) has invited three married couples to his summer house for a week: Bob(B) and Carrie (C) Hanson, David (D) and Edith (E) Irwin, and Frank (F) and Gena (G) Jackson. Since all six guests enjoy playing tennis match against every other guest except his/her spouse. In addition, Alvin is to play a match against each of David, Edith, Frank, and Gena. If no one is to play two matches on the same day, what is a schedule of matches over the smallest number of days.	08	4	4
8b	i. Is it possible to have a graph with the degree sequence 1,2,2,3,4,5? If yes, draw an example. If not, give reason. ii. Define the degree of balance of a signed graph. The following graph models the social relation between 4 people with '+' indicating friendship and '–' indicating enmity. Find the degree of balance of the graph.	08	3	3



9a	Find the extremals of the functional $\int_{\frac{1}{10}}^{1} y'\left(1+x^2y'\right)dx$, if $y\left(\frac{1}{10}\right)=19$, $y(1)=1$.	08	3	3
9b	A company needs to transport goods from its main warehouse (location A) to its retail outlet (location J). Along the way, the goods must pass through a series of distribution centres (B, C, D, E, F, G, H, I). At each segment of the journey, transport costs are incurred. The company wants to minimize the total transportation cost from A to J. Using dynamic programming, determine the minimum transportation cost from A to J and identify the optimal route. The transportation costs between centres are given in the table below:	08	2	2

	B	C	D
A	5	3	6

	E	F	G
B	4	2	5
C	3	4	6
D	5	3	2

	H	I
E	3	4
F	2	5
G	6	3

	J
H	4
I	3

OR

10a	Model the problem of a uniform chain (cable or rope) of length $2l$, suspended under gravity between two points at equal height. Find the curve when the chain is in equilibrium.	08	4	4
10b	Discuss any five optimization techniques and any three optimization principles.	08	2	2